

Handheld Digital Multimeters

Digital multimeters are the most widely used test and measurement equipment. Considering that they all look more or less the same, selecting the right multimeter for your application can be very confusing and difficult. Here is how to choose one



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There are hundreds of companies around the world manufacturing digital multimeters. Each of them offers several models with slightly different specifications. If judged visually, they all look alike with similar features. It becomes almost impossible to know which one is better than the rest for your requirements. Some of the cheap Chinese products look exactly the same as high-end popular multimeters but lack the same design and safety features. Therefore careful evaluation is a must to choose the best multimeter for your application.

Broadly, there are four main criteria for assessing a digital multimeter: measurement confidence, measure-

ment convenience, safety and construction.

Measurement confidence

For all engineering jobs like research, design and testing, measurements have to be correct because all your analysis will depend on the measured value. If a value is not measured correctly, you might end up with no or strange results. There is also a possibility that due to lack of understanding of the technical specification, you are taking these measurements with a wrong multimeter that does not support the range of measured quantity. So understanding technical specifications in the datasheet is a must to select a multimeter that will support the required range.

Technical specifications are not as complicated as they look. Here I explain these one by one:

Range. This specification defines the minimum and maximum of the quantities that can be measured without significant errors. This is the first thing that you need to check while selecting a multimeter. If you have to measure electrical lines up to 1000V but have a multimeter that can read a maximum of 600V, you will never get the results beyond 600V. If you are working on low-power circuits, you will have to read currents in microamperes. So you would require a multimeter that can read current in microamperes too.

Do check the range of other measurement functions like resistance,